FIZZAH AMIR

Computer Science Student — AI/ML Enthusiast

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PROFILE

Computer Science undergraduate with hands-on experience building end-to-end machine learning and deep learning systems — from classical ML pipelines (XGBoost, Random Forest) to CNN-based computer vision and Retrieval-Augmented Generation (RAG) systems with LLMs. Comfortable working across the full stack of an AI project: data collection and preprocessing, model training and evaluation, and deployment. Seeking an AI internship to contribute to real-world ML, generative AI, and computer vision projects.

EDUCATION

BS Computer Science, 4th Semester 2024 – 2028
Information Technology University (ITU), Lahore

FSc, Pre-Engineering Oct 2022 – Aug 2024
Kinnaird College for Women (KCW), Lahore

High School Diploma — Mathematics, Physics, CS Jul 2022 – 2024
Beaconhouse School System

TECHNICAL SKILLS

Languages: Python, C++

ML / Deep Learning: TensorFlow, Keras, Scikit-learn, XGBoost, CNN, GRU, Logistic Regression, Decision Tree, Random Forest, SMOTE, Feature Engineering, Time-Series Analysis

Generative AI / LLMs: Hugging Face Sentence Transformers, RAG Pipelines, LLM Prompting, FAISS (vector search)

Computer Vision: OpenCV, scikit-image, Optical Flow (Farneback), Image/Video Preprocessing

Data & Databases: PostgreSQL, Oracle Database, SQL, Pandas, NumPy

Backend & APIs: Django, Django REST Framework, Node.js (learning), Express.js, REST APIs, JWT Auth, RBAC

Mobile: Flutter, Dart

Cloud & Tools: AWS EC2, Railway Cloud, Git, GitHub, Linux

CS Core: OOP, DSA, Algorithms, Network Programming

AI / ML PROJECTS

Pakistan Sign Language (PSL) Recognition — Real-Time Sign-to-Speech System *Ongoing · ITU*

- Building a real-time system where a webcam captures PSL hand signs and converts them into Urdu text and speech, using a Live Webcam → Optical Flow → CNN → Text → Speech pipeline.
- Collected and organized two datasets (5.77 GB, 23,499 files): UAlpha40 (36 static PSL alphabet signs + 4 dynamic letters) and the PSL Dictionary Toolkit (80 word-level signs scraped from psl.org.pk).
- Built a preprocessing pipeline that samples 16 frames per video and computes dense optical flow (Farneback algorithm) between consecutive frames to capture motion signatures, producing (15, 64, 64, 2) motion-map arrays per sample.
- Implemented data augmentation (flipping, brightness adjustment, Gaussian noise) to expand limited dynamic-sign samples, and visually verified optical flow output via HSV motion maps.
- Currently developing a CNN+GRU architecture to classify dynamic signs from optical-flow sequences, with an end-to-end app to translate recognized signs to Urdu text and speech.

Tech: *Python · TensorFlow · Keras · OpenCV · scikit-image · NumPy · MoviePy · deep-translator · gTTS · Computer Vision · CNN · GRU*

Electricity Theft Detection Using Machine Learning *ITU | Group Project*

- Built an end-to-end ML pipeline to detect electricity theft from raw smart meter time-series data (UCI dataset, 2M+ records).
- Performed full data preprocessing: missing-value handling via forward-fill and interpolation, noise removal, normalization, and daily/weekly aggregation.
- Engineered features including rolling average consumption, peak vs. off-peak ratios, consumption trends, variance, and hour/day-wise behavior patterns.

- Trained and benchmarked classifiers — Logistic Regression, Decision Tree, Random Forest, XGBoost, and a TensorFlow-based neural network — addressing class imbalance using SMOTE.
- Produced alerts, classification reports, and consumption-pattern visualizations as post-processing output.

Tech: *Python · TensorFlow · Scikit-learn · XGBoost · Pandas · NumPy · SMOTE · Matplotlib*

Meeting Transcriber & Q&A Assistant (RAG) *Personal Project*

- Built a meeting transcription and question-answering tool converting recorded audio into searchable, queryable knowledge via a Retrieval-Augmented Generation (RAG) pipeline.
- Transcribed audio to text, chunked transcripts, and generated dense vector embeddings using a Hugging Face Sentence Transformer (all-MiniLM-L6-v2) for semantic search.
- Indexed embeddings in FAISS to retrieve the most relevant transcript chunks per query using cosine similarity.
- Implemented a RAG pipeline feeding retrieved context into an LLM prompt for grounded, citation-backed natural-language answers, plus automatic summarization and action-item extraction.

Tech: *Python · Hugging Face Sentence Transformers · FAISS · RAG · LLM Prompting · Speech-to-Text*

OTHER SOFTWARE ENGINEERING PROJECTS

Digital Wallet System API *Live on AWS EC2*

- Built a complete digital wallet REST API (similar to EasyPaisa/JazzCash) with Django REST Framework and PostgreSQL, deployed on AWS EC2.
- Implemented JWT authentication with custom claims, Argon2 password hashing, and Role-Based Access Control across 3 roles.
- Designed 29 REST API endpoints using raw SQL with explicit transactions and parameterized queries; wrote 8 PostgreSQL triggers and a 5-level automatic rank system.

Tech: *Python · Django · Django REST Framework · PostgreSQL · JWT · AWS EC2 · Argon2*

AeroHub Management System *Dec 2025 – Jan 2026 · ITU*

- Full airport management system in C++ with a custom HTTP server, deployed on Railway Cloud with an HTML/CSS/JS frontend.
- Implemented 6 data structures from scratch: B-Tree, Dijkstra's algorithm on a graph, HashMap, Linked List, Priority Queue, and Bitmap.

Tech: *C++ · Custom HTTP Server · HTML/CSS/JS · REST API · Railway Cloud · File I/O*

In-Memory Cache Server — Redis Clone *ITU | Advanced Database Management*

- Built a single-threaded in-memory key-value server in C++ implementing the Redis wire protocol (RESP) with an event-driven epoll core.
- Built four core data types from scratch and added crash-safe persistence; sustained 50,000+ SET/GET ops/sec in benchmarks.

Tech: *C++ · epoll · TCP Sockets · RESP Protocol · Skip Lists · Hash Tables · Systems Programming*

Flight Management System *Jun 2024 – Aug 2024 · ITU*

- Console-based C++ system with flight scheduling, seat selection, reservation management, and pilot assignment using core OOP principles and the Singleton pattern.

Tech: *C++ · OOP · File I/O · Singleton Pattern*

Color Detection App for Color Blind Users *Personal Project*

- Flutter mobile app enabling color blind users to identify colors in real time with an accessible interface.

Tech: *Flutter · Dart · Mobile Development*

LICENSES & CERTIFICATIONS

Foundations of Data Science Issued Jun 2026

Google · Coursera — Credential ID VAQAXU7QH6MU

Cybersecurity Bootcamp: ISO 27001:2022 Issued April 2025

EXTRACURRICULAR & COMPETITIONS

Competitive Programming — CodeRush May 2025

Competed in algorithmic problem-solving challenges, applying data structures and algorithms under time constraints.

Competitive Programming — SOFTEC 2026

Solved algorithmic and data structure problems in a competitive environment.